NAHIDA ZAMAN BINA

Data Scientist • Analytics Engineer

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github.com/nahida30



TECHNICAL SKILLS

Data Science & ML	Scikit-learn, XGBoost, Predictive Modeling, Feature Engineering, Stacking Ensembles, Pandas, NumPy, Matplotlib
Data Architecture & SQL	Relational Databases, Complex SQL Queries, MySQL, Data Modeling Techniques, Schema Design
Automation & Scripting	Python (Primary), Linux Shell Scripting basics, RESTful API Integration, Workflow Orchestration concepts
Deep Learning & NLP	BanglaBERT, Hugging Face Transformers, LangChain, FAISS Vector Stores, Computer Vision (CNNs)
Mathematics	Probability & Statistics, Linear Algebra, Data Structures & Algorithms

DATA SCIENCE & ANALYTICS PROJECTS

Customer Churn Prediction & Behavioral Analysis | https://github.com/nahida30/Customer-Churn-Prediction

- Analyzed customer lifecycle datasets, usage behaviors, and retention risk patterns to identify key factors leading to user churn.
- Engineered a predictive pipeline utilizing machine learning classifiers to isolate high-risk customer segments, enabling proactive marketing mitigation strategies.
- Evaluated model performance using statistical metrics (Precision, Recall, ROC-AUC) to optimize classification accuracy and ensure reliable business insights.

Credit Card Fraud & Risk Analysis | https://github.com/nahida30/Credit-Card-Fraud-Risk-Analysis

- Developed an end-to-end data analysis and predictive modeling framework to detect fraudulent financial transactions and mitigate risk.
- Applied advanced feature engineering, handled extreme class imbalances, and optimized classification algorithms (XGBoost/Scikit-learn) for high-volume, high-sensitivity data.
- Translated model outputs into actionable reporting metrics, aligning statistical data with operational risk management frameworks.

Crop Recommendation System | *https://github.com/nahida30/Crop-Recommendation-System*

- Built a predictive multi-class classification model using environmental and soil metrics to automate optimized decision-making for agricultural yield.
- Conducted exploratory data analysis (EDA) and applied statistical concepts to ensure dataset alignment with predictive modeling requirements.

Bangla PDF RAG System & Next Word Prediction | *https://github.com/nahida30/bangla-rag*

- Designed an end-to-end Retrieval-Augmented Generation (RAG) pipeline utilizing LangChain, FAISS vector databases, and Groq LLM APIs for automated document querying.

- Fine-tuned a custom BanglaBERT model from scratch using the Hugging Face Trainer API, executing custom tokenization and text preprocessing.

PROFESSIONAL EXPERIENCE

Junior Software Engineer | Revenco | *Dec 2023 – Mar 2025* | Dhaka, Bangladesh

- Managed backend logic and database schema architectures using MySQL, writing optimized, complex SQL queries to manage high-volume application data.
- Designed and integrated production-grade RESTful APIs for database-backed systems, ensuring efficient data processing and seamless workflow connectivity.
- Collaborated within cross-functional teams to translate business requirements into application logic, building structured data models for platforms like Tizaraa (e-commerce).
- Utilized Python and JavaScript to automate backend scripting tasks, improving data transit and processing efficiency.

Content & Technical Specialist | Green Earth Resilience Foundation | *Jan 2024 – Jul 2024* | Remote

- Synthesized research-backed technical datasets for sustainability platforms, translating highly complex statistical and scientific findings into clear executive insights.
- Managed cross-functional content frameworks with an emphasis on rigorous analytical validation and structure.

RESEARCH & ML PROJECTS

Early Prediction of Sepsis — Python, Scikit-learn, XGBoost, ANN, MLP

Implemented and compared multiple ML models and a hybrid stacking ensemble for clinical Sepsis prediction; achieved high classification accuracy via cross-validation and feature engineering.

Next Word Prediction in Bangla — Python, Hugging Face Transformers, BanglaBERT

Trained a custom BanglaBERT model from scratch for Masked Language Modeling using custom WordPiece tokenization and BertForMaskedLM + Trainer API.

PUBLICATION

"Palm Leaf Health Management: A Hybrid Approach for Automated Disease Detection and Therapy Enhancement" — *IEEE Access, Vol. 12*

Authors: S. M. Nuruzzaman Nobel, Md. Asif Imran, Nahida Zaman Bina, Md. Mohsin Kabir, Mejdl Safran, Sultan Alfarhood, M. F. Mridha

EDUCATION

M.Sc. Engg. in ICT — **CGPA: 4.00 / 4.00** — Dec 2024 – Running
Bangladesh University of Professionals (BUP)

B.Sc. Engg. in Computer Science & Engineering — **CGPA: 3.69 / 4.00** — Feb 2020 – Jan 2024
Bangladesh University of Business & Technology (BUBT)

CERTIFICATIONS & ACHIEVEMENTS

- Black Belt Certificate — Web Development Course, Programming Hero (2023)
- Data Analytic Course, Ostad (2026)